

Needs-Based Recommendation System

Estimating Financial Needs
with Machine Learning

Business Case 2 | ML Fintech | Politecnico di Milano

5,000
real clients

11
financial products

2
investment needs predicted

Our Approach

End-to-end pipeline from raw data to personalized product recommendations



EDA

Life-cycle
validation



**Expert
Picking**

Bias
filtering



Modeling

7 algorithms
compared



XAI

SHAP +
surrogate tree



Fairness

Gender &
Age audit



NBA

Hybrid
recommender

7 features

Age, Gender, Family,
FinEdu, Risk, Income, Wealth

2 targets

IncomeInvestment
AccumulationInvestment

11 products

Risk range: 0.12 – 0.88
Implicit labels

EDA: Key Findings

Life-Cycle Validation

Income need follows age perfectly:

< 35 years:	17%	probability
35-50:	26%	
50-65:	33%	
65+:	76%	

Chi-square test: $p < 0.001$

Accumulation is ~50% across all ages — age-independent, driven by income/wealth instead.

Class Balance

P(Income=1) = 38.4%

mild imbalance → threshold tuning needed

P(Accumulation=1) = 51.3%

naturally balanced

Normalization Strategy

Shapiro-Wilk test → all features non-normal

→ **MinMaxScaler for all**

Wealth & Income: log-transformed first to reduce fat-tail skewness

One-vs-All vs Multi-label

Should we model both needs jointly or independently?

One-vs-All (chosen)

2 independent binary classifiers

(1 per need)

- Simpler, more robust
- Easier to tune per-target
- Occam's Razor: fewer parameters

F1 Income: 0.642 F1 Accum: 0.776

ClassifierChain (tested)

1 chained model for both targets

(prediction of one feeds the other)

- Can capture cross-need correlations
- More complex, risk of error propagation

F1 Income: 0.642 F1 Accum: 0.772

Target correlation (ϕ) = 0.011 → **near zero. No benefit from joint modeling. One-vs-All wins by Occam's Razor.**

Explainability

SHAP Values

Income: Age is the #1 driver — confirms life-cycle theory

Accumulation: Income_log + Wealth_log dominate

Feature importance and SHAP align — reinforcing trust in the model

Surrogate Tree

Depth-4 tree mimicking XGBoost

Income fidelity: 85%

Split: Wealth_log $\leq 0.56 \rightarrow$ Age

Accumulation fidelity: 76.5%

Split: Income_log ≤ 0.55

Actionable Rules for Advisors

Income: Wealth > median AND Age > 55 $\rightarrow$ high need probability

Accumulation: Income < median $\rightarrow$ high need

Fairness Audit

EU AI Act compliance check

Gender

Disparate Impact = 0.999

Equal Opportunity gap = 0.002

Selection rate: 27.7% for both genders

Fully fair ✓

Age

DI Ratio (Income) = 0.129 — below 0.80

But: reflects real life-cycle patterns,
not algorithmic bias.

Income Selection Rate by Age

Age	Sel. Rate	TPR
< 35	11.1%	0.250
35 – 50	10.2%	0.187
50 – 65	19.1%	0.400
65+	79.2%	0.937

The disparity in age is not bias — older clients genuinely have higher income investment needs, as validated by the life-cycle analysis.

Ensemble Strategy

From individual models to combined intelligence

Soft Voting

Average probabilities

Equal-weight average of SVM, XGBoost, and Random Forest predicted probabilities.

Best for Accumulation:

Acc: 0.787 - F1: 0.801

Stacking

Learned combination

Logistic Regression meta-learner trained on cross-validated base model outputs.

Discovers which model is best for which client segment.

Ensemble Uncertainty Filter

Credibility intervals

Method: Empirical Standard Deviation

*Calculates the disagreement (std) between SVM, XGB, and RF. If the disagreement is high (> 0.15), the system triggers an **abstention** to ensure MiFID compliance. **Benefit:***

Isolates borderline cases for human review, significantly increasing F1-score on the remaining "active" clients.

Voting wins over Stacking: base models agree enough that learned weighting adds no value — simplicity prevails.

Model Comparison

Income Investment

Model	ACCURACY	F1
SVM	0.733	0.652
XGBoost	0.763	0.641
Random Forest	0.770	0.639
MLP Neural Net	0.759	0.606
Soft Voting	0.768	0.644
Stacking	0.771	0.640

Accumulation Investment

Model	ACCURACY	F1
SVM	0.684	0.731
XGBoost	0.768	0.770
Random Forest	0.779	0.780
MLP Neural Net	0.770	0.75
Soft Voting	0.787	0.801
Stacking	0.790	0.794



Soft Voting: best for Accumulation (ACC 0.787, F1 0.801). SVM remains competitive — simple models work well here. Threshold tuning (optimal $\neq$ 0.5) further improves F1.

Recommendation Engine

01 Need Detection

ML classifier predicts Income / Accumulation need per client.

Threshold optimized via Expected Monetary Value analysis.

02 Risk Matching

Content-based filter: match client RiskPropensity with product risk level.

Kernel-based similarity for smooth ranking.

03 Hybrid SVD

Collaborative filtering via TruncatedSVD on synthetic user-item matrix.

50/50 blend with content-based. 99% variance in $k=5$.

Product Gap Analysis: 17.5% clients underserved → catalog expansion proposals with estimated profit delta

Product Catalog & Gap Analysis

Product Catalog (11 products)

Product	Type	Risk
Balanced Mutual Fund	Accum	0.55
Income Conservative UL	Income	0.30
Fixed Income MF	Income	0.12
Balanced High Div MF	Income	0.44
Balanced Mutual Fund	Accum	0.41
Defensive Flex UL	Accum	0.36
Aggressive Flex UL	Accum	0.75
Balanced Flex UL	Accum	0.48
Cautious Alloc SA	Accum	0.27
Fixed Income SA	Income	0.13
Aggressive Alloc SA	Accum	0.88

Gap Detection Results

17.5% of clients (877) have
match quality < 0.75

Decision Tree identifies 3 underserved segments:

Risk < 0.11: 186 very conservative clients

Risk 0.23–0.29: 519 moderate clients

Risk 0.60–0.74: 172 growth-oriented clients

→ 3 new products proposed to fill these gaps

Business Impact

Expected Monetary Value: should we optimize for precision or recall?

R = 100€ revenue per conversion

C = 40€ cost per contact

L = 50€ lost opportunity

Income Investment

Threshold (F1 optimal) **0.370**

Threshold (EMV optimal) **0.190**

Profit leak at F1 thr. **€4,080**

Business orientation **RECALL**

Accumulation Investment

Threshold (F1 optimal) **0.460**

Threshold (EMV optimal) **0.370**

Profit leak at F1 thr. **€1,050**

Business orientation **RECALL**

Both targets are recall-oriented: missing a real need costs more than a wasted contact.

New Product Proposals

Three products to fill the underserved segments identified by the Gap Analysis

Conservative Accumulation

Risk level: 0.07

Target segment: 186 clients

Median client risk: 0.094

No existing product below 0.13 — catalog dead-zone.

+ €16,933

expected profit uplift

Moderate Accumulation

Risk level: 0.19

Target segment: 1,092 clients

Median client risk: 0.272

Fills gap between Cautious Alloc SA (0.27) and lower tiers.

Volume driver

largest underserved segment

Moderate Income

Risk level: 0.23

Target segment: 584 clients

Median client risk: 0.325

Fills gap between Income products at 0.13 and 0.30.

Cross-sell

widens the Income catalog

Accumulation launch decision: LAUNCH RECOMMENDED — profit +€7,778

Conclusions

What We Achieved

- End-to-end pipeline: data → recommendations
- Best model: F1 0.801 (Accumulation)
- Full XAI: SHAP + interpretable surrogate tree
- Fairness: gender-fair, age explained
- Hybrid recommender with Business Logic Layer (Conflict Resolution & Risk Matching)
- EMV threshold: recall-oriented for both

Thank you!

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